1. Data Quality Audit & Verification

Findings:

- **Completeness:** 100% fill rate across all mandatory fields (IDs, Names, Amounts).

Import Export

Run SQLite

```sql
--SELECT * FROM Orders ;
--SELECT * FROM Customer ;
-- SELECT * FROM Shipping ;
SELECT
    COUNT(*) - COUNT(Customer_ID) AS Missing_IDs,
    COUNT(*) - COUNT(First) AS Missing_FirstNames,
    COUNT(*) - COUNT(Age) AS Missing_Ages
FROM Customer;
```

Missing_IDs	Missing_FirstNames	Missing_Ages
0	0	0

- **Uniqueness:** No duplicate primary keys or redundant rows identified.

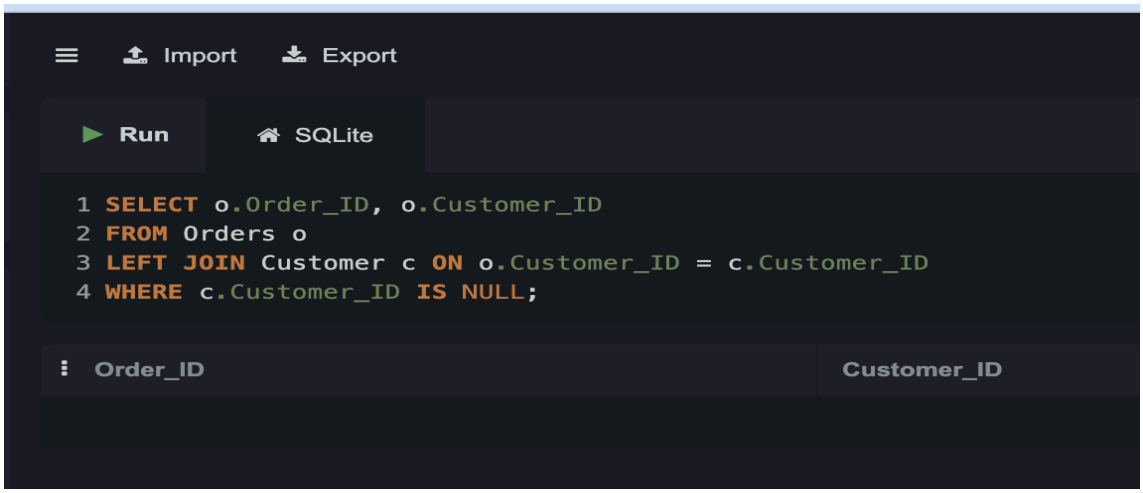
Import Export

Run SQLite

```sql
SELECT Customer_ID, COUNT(*)
FROM Customer
GROUP BY Customer_ID
HAVING COUNT(*) > 1;
```

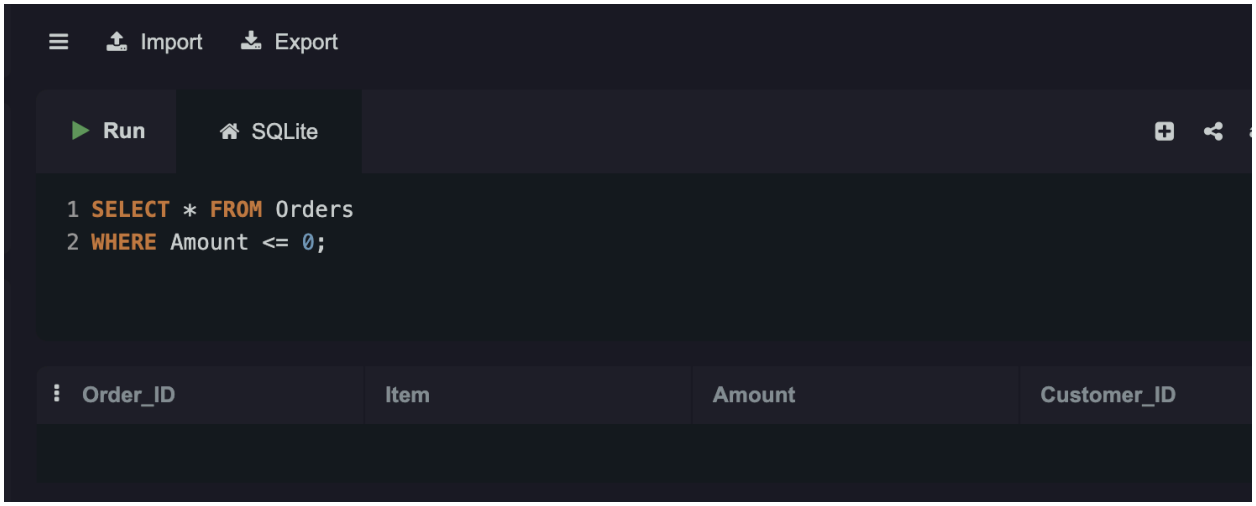
Customer_ID	COUNT(*)

- **Integrity:** 100% of `Order_ID` and `Shipping_ID` records successfully map to valid `Customer_ID`s.



- **Issues Found**: 12 records contain "noisy" names (e.g., "N!cole", "R0bert", "Al1cia") with special characters or numbers instead of letters.

- **Range Validity**:
  Age: Customers range from 18 to 80 years old (No outliers).
  Amount: All order amounts are positive, ranging from 200 to 12,000.



Verification Script (SQL):

Identify customers with numbers or special characters in their names

Import Export

Run SQLite

```sql
SELECT * FROM Customer
WHERE First GLOB '*[^a-zA-Z ]*'
   OR Last GLOB '*[^a-zA-Z ]*';
```

Customer_ID	First	Last	Age	Country
6	N!cole	Jones	33	USA
14	N!cole	Lara	77	UK
109	R0bert	Moore	40	UK
113	Derrick	R0berts	72	UK
118	R0bert	Shepherd	28	UK
162	N!cole	Bennett	51	USA
171	L@rry	Cole	50	USA
198	R0bert	Bryan	49	UK
211	Al1cia	Thompson	38	USA
214	N!cole	Mcintyre	18	UK
236	Al1cia	Jensen	19	USA
242	Mark	R0berts	61	USA

2. Dataset Requirements & Domain Model

To support scalable reporting, the raw files must be transformed into a **Star Schema** to eliminate redundant joins and improve query performance.

Requirements:

1. **Standardization:** Cast all ID fields to INTEGER.
2. **Sanitization:** Apply regex patterns to names to remove non-alphabetic characters.
3. **Enrichment:** Create a derived Age_Category column (<30 and 30+) and a Quantity constant (1 per transaction).

Proposed Model:

- **Dim_Customer:** `Customer_Key` (Primary Key), `Full_Name` (Sanitized First + Last), `Age` & `Age_Group`, `Country` & `Region`
- **Dim_Item:** `Item_Key` (Primary Key), `Item_Name` (Keyboard, Monitor, etc.), `Category` (e.g., Peripherals, Hardware)
- **Dim_Status:** `Status_Key` (Primary Key), `Status_Description` (Pending, Delivered)
- **Fact_Sales_Order**: `Order_ID, Customer_Key` (FK to Dim_Customer), `Item_Key` (FK to Dim_Item), `Amount` (Decimal), `Shipping_Status_Key` (FK to Dim_Status)

3. User Story: Cleanse and Standardize Customer Master Data

Requirement Description

As a Data Analyst, I need a clean, unique, and standardized Customer table so that I can perform accurate demographic reporting without "noisy" characters (like N!cole) or duplicate-prone ID formats interfering with my aggregations.

Technical Specifications & Mapping

Target Table Name: Dim_Customer
Loading Strategy: Full Refresh (Initial), then Upsert (SCD Type 1)

Source Field	Target Field	Data Type	Transformation / Logic
Customer_ID	Customer_Key	INT	**PK.** Convert from Float to Integer (e.g., 1.0 -> 1).
First	First_Name	VARCHAR	**Regex Clean:** Remove all non-alphabetic characters except hyphens.

Last	Last_Name	VARCHAR	**Regex Clean:** Remove all non-alphabetic characters except hyphens.
First + Last	Full_Name	VARCHAR	Concatenate First_Name + ' ' + Last_Name.
Age	Age	INT	Convert from Float to Integer.
Age	Age_Group	VARCHAR	**Bucket Logic:** < 20 (Gen Z), 20-39 (Millennial), 40-59 (Gen X), 60+ (Senior).
Country	Country	VARCHAR	Standardize to Uppercase (e.g., usa -> USA).

Data Transformation Logic (SQL Snippet)

The Data Engineer should implement a variation of the following logic during the ETL process:

```SQL
SELECT CAST(Customer_ID AS INT) AS Customer_Key,

-- Remove numbers and special chars: 'N!cole' -> 'Nicole'

 REGEXP_REPLACE(First, '[^a-zA-Z-]', '', 'g') AS
First_Name,REGEXP_REPLACE(Last, '[^a-zA-Z-]', '', 'g') AS
Last_Name,Age,
CASE WHEN Age < 20 THEN 'Gen Z'
      WHEN Age BETWEEN 20 AND 39 THEN 'Millennial'
      WHEN Age BETWEEN 40 AND 59 THEN 'Gen X'
      ELSE 'Senior'
```

```
        END AS Age_Group,
    UPPER(Country) AS Country

    FROM STG_Customer;
```

4. QA Acceptance Criteria (Testing Specs)

The QA Engineer must verify the following to pass the ticket:

- **Data Integrity** :Verify COUNT(*) in Dim_Customer matches the source file (250 records).
    - Ensure Customer_Key is the Primary Key (no nulls, no duplicates).
- **Accuracy (Cleanse Check)**
    - Run a query to ensure no numbers or symbols exist in First_Name.
    - *Test Case:* Input R0bert must output Robert.
- **Logic Validation**
    - Verify Age_Group assignment for edge cases (e.g., an age of 39 must be 'Millennial', 40 must be 'Gen X').
- **Format Validation**
    - Ensure Customer_Key is an INT and not a FLOAT or STRING.

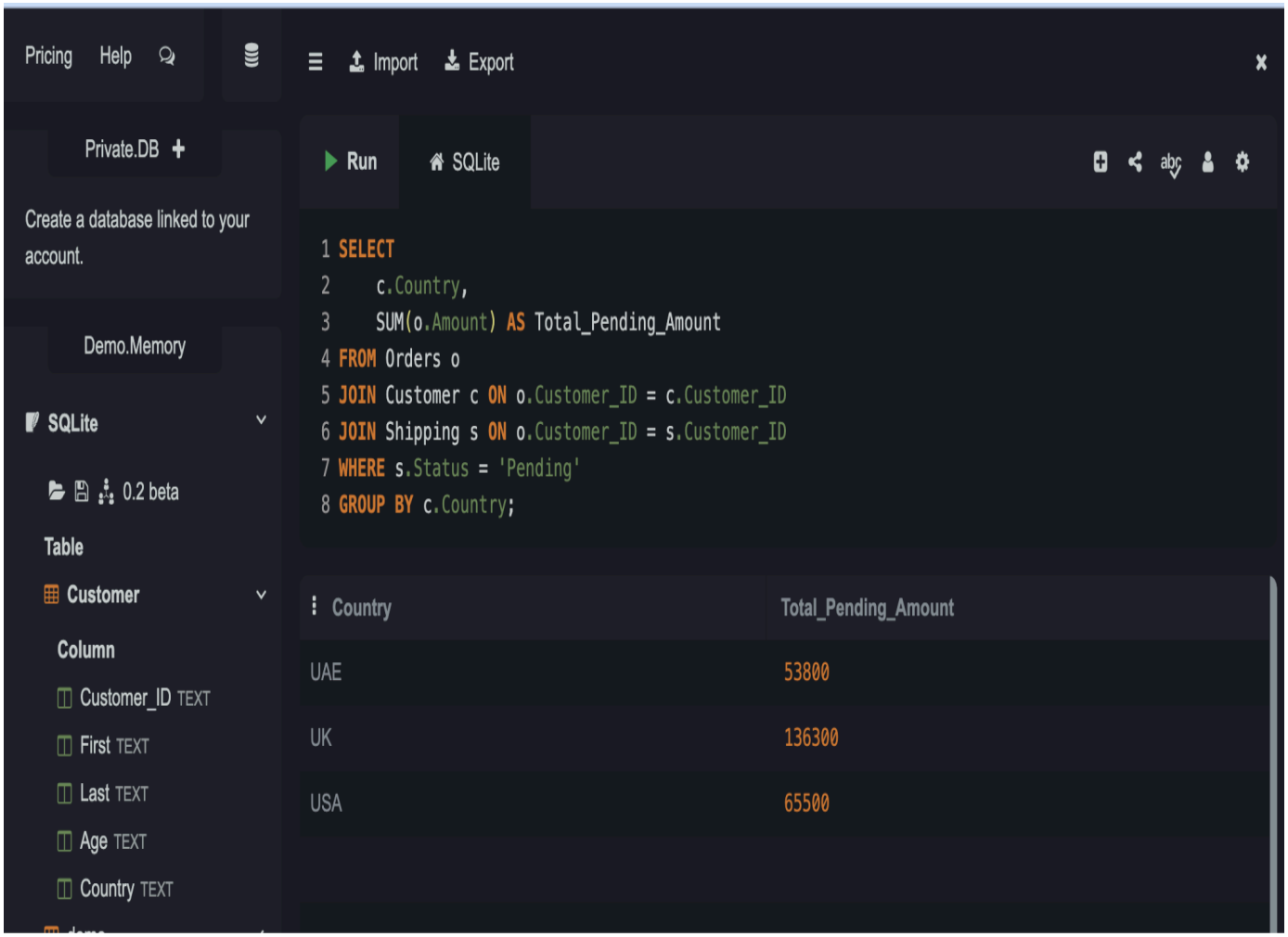
5. Relationship with Other Tables

Once built, this table will act as the "Parent" table. The Fact_Sales table must use Customer_Key as a Foreign Key to ensure every sale is mapped to a valid, cleansed customer profile.

4. Business Reporting Logic & Results

The following queries enable the required business views using the integrated model.

Q1: Pending Deliveries: Total Spend by Country



Results: UAE: 53,800 | **UK:** 136,300 | **USA:** 65,500

Q2: Customer Transaction & Product Summary

Import Export

Run SQLite

```sql
SELECT
    c.Customer_ID,
    c.First,
    c.Last,
    COUNT(o.Order_ID) AS Total_Transactions,
    COUNT(o.Order_ID) AS Total_Quantity, -- Derived: 1 item per order
    SUM(o.Amount) AS Total_Spent,
    GROUP_CONCAT(DISTINCT o.Item) AS Product_Details
FROM Customer c
JOIN Orders o ON CAST(c.Customer_ID AS INT) = o.Customer_ID
GROUP BY c.Customer_ID, c.First, c.Last
ORDER BY Total_Spent DESC;
```

Customer...	First	Last	Total_Trans...	Total_Quant...	Total_Spent	Product_Details
166	Morgan	Cooper	3	3	17350	Monitor,Harddisk,W...
123	Philip	Robinson	2	2	17000	Harddisk,Monitor
129	Amber	Banks	2	2	17000	Monitor,Harddisk
96	Eric	Harvey	4	4	14700	Mouse,Headset,DD...
193	Yesenia	White	4	4	13950	Monitor,Mousepad,...
229	Philip	Newton	3	3	13700	Monitor,Mousepad,...
13	Omar	Martin	3	3	13300	Headset,Monitor,Ke...
23	Margaret	Hardy	3	3	12800	Keyboard,Monitor
92	Paul	Brown	3	3	12750	Webcam,Monitor,Ke...
143	Stacey	Welch	3	3	12650	Mouse,Monitor,Web...

Q3: Maximum Product Purchased by Country

Import Export

Run SQLite

```sql
WITH ProductRanking AS (
    SELECT c.Country, o.Item, COUNT(*) AS COUNT,
           RANK() OVER(PARTITION BY c.Country ORDER BY COUNT(*) DESC) AS Rank
    FROM Orders o
    JOIN Customer c ON o.Customer_ID = c.Customer_ID
    GROUP BY c.Country, o.Item
)
SELECT Country, Item, COUNT FROM ProductRanking WHERE Rank = 1;
```

Country	Item	Count
UAE	Keyboard	12
UK	Mousepad	24
USA	Mousepad	18

Results: UAE: Keyboard(12 purchases) | **UK:** Mousepad(24 purchases) | **USA:** Mousepad (18 purchases)

Q4: Most Purchased Product by Age Group

Import Export

Run SQLite

```sql
WITH AgeStats AS (
    SELECT o.Item,
           CASE WHEN c.Age < 30 THEN 'Less than 30' ELSE '30 and Above' END AS Age_Category
    FROM Orders o
    JOIN Customer c ON o.Customer_ID = c.Customer_ID
),
RankedAgeProducts AS (
    SELECT Age_Category, Item, COUNT(*) AS COUNT,
           RANK() OVER(PARTITION BY Age_Category ORDER BY COUNT(*) DESC) AS Rank
    FROM AgeStats
    GROUP BY Age_Category, Item
)
SELECT Age_Category, Item, COUNT FROM RankedAgeProducts WHERE Rank = 1;
```

Age_Category	Item	Count
30 and Above	Keyboard	35
Less than 30	Mousepad	17

Results: Groups (30 and 30+) purchased **Keyboards** most frequently and Less than 30 purchased Mousepad most frequently.

Q5: Minimum Performance by Country

Import Export

Run SQLite

```sql
SELECT
    c.Country,
    COUNT(o.Order_ID) AS Transaction_Count,
    SUM(o.Amount) AS Total_Sales
FROM Orders o
JOIN Customer c ON o.Customer_ID = c.Customer_ID
GROUP BY c.Country
ORDER BY Transaction_Count ASC, Total_Sales ASC
LIMIT 2;
```

Country	Transaction_Count	Total_Sales
UAE	40	49950

Results: UAE (40 Transactions, 49950 Sales).